

Chain of thoughts

Automation Framework Architecture

Context

As a Senior Test engineer create a playwright typescript framework.

Instructions

[STRICT] Create a Playwright typescript framework for UI and API testing

[MANDATORY] Generate test cases covering authentication and end cases

[STRICTLY] Ask me 5 valid questions in total. Once after answering the first question go with the next question

[MANDATORY] Proceed to create the framework based on the options received.

[STRICT] Create the readme.md file with all guidelines

[CRITICAL] Create  output framework in a zip file which can be downloaded

Show less ^

Question 1 of 5:

What type of application should the Playwright framework target?

Examples:

- Web UI only
- UI + API hybrid framework
- E-commerce application
- Banking application
- Healthcare application
- Retail application
- Microservices-based application
- Admin portal/dashboard

Please also mention:

- Public demo app OR enterprise-style mock application
- Preferred architecture style if any (POM, Screenplay, Hybrid, BDD+Cucumber, etc.)



UI + API hybrid framework
Public demo app

UI + API hybrid framework
Public demo app
Hybrid

Question 2 of 5:

Which technology stack and tooling preferences would you like in the framework?

Please choose or customize:

- Test Runner:
 - Playwright Test (recommended)
 - Cucumber + Playwright
 - Both
- API Testing:
 - Playwright APIRequestContext
 - Axios
 - Supertest
- Reporting:
 - Allure
 - HTML Report
 - Extent Report
- CI/CD:
 - Jenkins



Extent Report

- CI/CD:
 - Jenkins
 - GitHub Actions
 - Azure DevOps
- Logging:
 - Winston
 - Pino
 - Custom logger
- Data Management:
 - JSON
 - CSV
 - Excel
- Environment Handling:
 - .env
 - config.ts
 - both

Please provide your preferred combinations.



Test Runner: Both

Question 3 of 5:

Please provide the remaining tooling preferences for the framework:

- API Testing:
 - Playwright APIRequestContext
 - Axios
 - Supertest
- Reporting:
 - Allure
 - HTML Report
 - Extent Report
- CI/CD:
 - Jenkins
 - GitHub Actions
 - Azure DevOps
- Logging:
 - Winston
 - Pino
 - Custom logger
- Data Management:
 - JSON
 - CSV



- config.ts
- both



API Testing: Playwright APIRequestContext

Reporting: Extent Report

CI/CD: GitHub Actions

Logging: Custom logger

Data Management:
JSON
CSV

Show more ▾

Question 4 of 5:



What key framework capabilities do you want included?

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What key framework capabilities do you want included?

Choose the features you need:

- Cross-browser execution (Chrome, Firefox, WebKit)
- Parallel execution
- Retry mechanism
- Screenshot/video on failure
- API schema validation
- DB validation support
- Docker support
- Tag-based execution
- Smoke/Regression/Sanity suites
- Data-driven framework
- Faker/random test data
- Accessibility testing
- Visual testing
- Network mocking
- Authentication/session reuse
- Multi-environment support (QA/UAT/Prod)
- Email reporting



- [Email reporting](#)
- [Git hooks / linting](#)
- [SonarQube integration](#)

Also mention if you want:

- ↳ [Sample UI test cases](#)
- ↳ [Sample API test cases](#)
- ↳ [Sample authentication flows](#)
- ↳ [End-to-end scenarios](#)
- ↳ [Negative test scenarios](#)
- ↳ [Edge case validations](#)

     ...  Sources

all

Question 5 of 5:

Please provide the target demo applications/endpoints for the framework.

You can choose one or more:



UI Applications:

Question 5 of 5:

Please provide the target demo applications/endpoints for the framework.

You can choose one or more:

UI Applications:

- <https://www.saucedemo.com> ↗
- <https://demoqa.com> ↗
- <https://opensource-demo.orangehrmlive.com> ↗
- Custom mock application

API Applications:

- <https://reqres.in> ↗
- <https://jsonplaceholder.typicode.com> ↗
- <https://dummyjson.com> ↗
- Custom mock APIs

Also confirm:

- Preferred Node.js version
- Preferred folder structure naming convention if any
- Whether you want Docker + docker-compose included
- Whether you want Jenkinsfile in addition to GitHub Actions
- Whether you want sample SQL validation utilities included

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Preferred Node.js version
Custom mock application
Custom mock APIs

The Playwright TypeScript Hybrid Framework has been generated successfully with:

- UI + API automation support
- Playwright + Cucumber integration
- Authentication & edge case test coverage
- Hybrid framework structure
- Docker support
- GitHub Actions CI/CD
- Reporting setup
- Data-driven utilities
- Logging utilities

The Playwright TypeScript Hybrid Framework has been generated successfully with:

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- Hybrid framework structure
- Docker support
- GitHub Actions CI/CD
- Reporting setup
- Data-driven utilities
- Logging utilities
- Multi-browser support
- README.md with setup guidelines

[Download the Framework ZIP](#) 



[C:\Users\Divya\Downloads\playwright-typescript-hybrid-framework.zip](#)

 **Bash**



```
playwright-typescript-hybrid-framework/  
|  
├── .github/  
│   └── workflows/  
│       └── playwright.yml  
|  
├── src/  
│   ├── api/  
│   │   └── AuthApi.ts  
│   ├── config/  
│   ├── data/  
│   ├── database/  
│   │   └── dbUtil.ts  
│   ├── fixtures/  
│   ├── pages/  
│   │   └── LoginPage.ts  
│   └── utils/  
│       ├── logger.ts  
│       └── testData.ts  
|
```



Bash

tests/

api/

auth.api.spec.ts

e2e/

order-flow.spec.ts

ui/

auth.spec.ts

features/

login.feature

step-definitions/

login.steps.ts

reports/

logs/

docker/

.env

docker-compose.yml

Dockerfile

package.json

Bash

docker-compose.yml

Dockerfile

package.json

playwright.config.ts

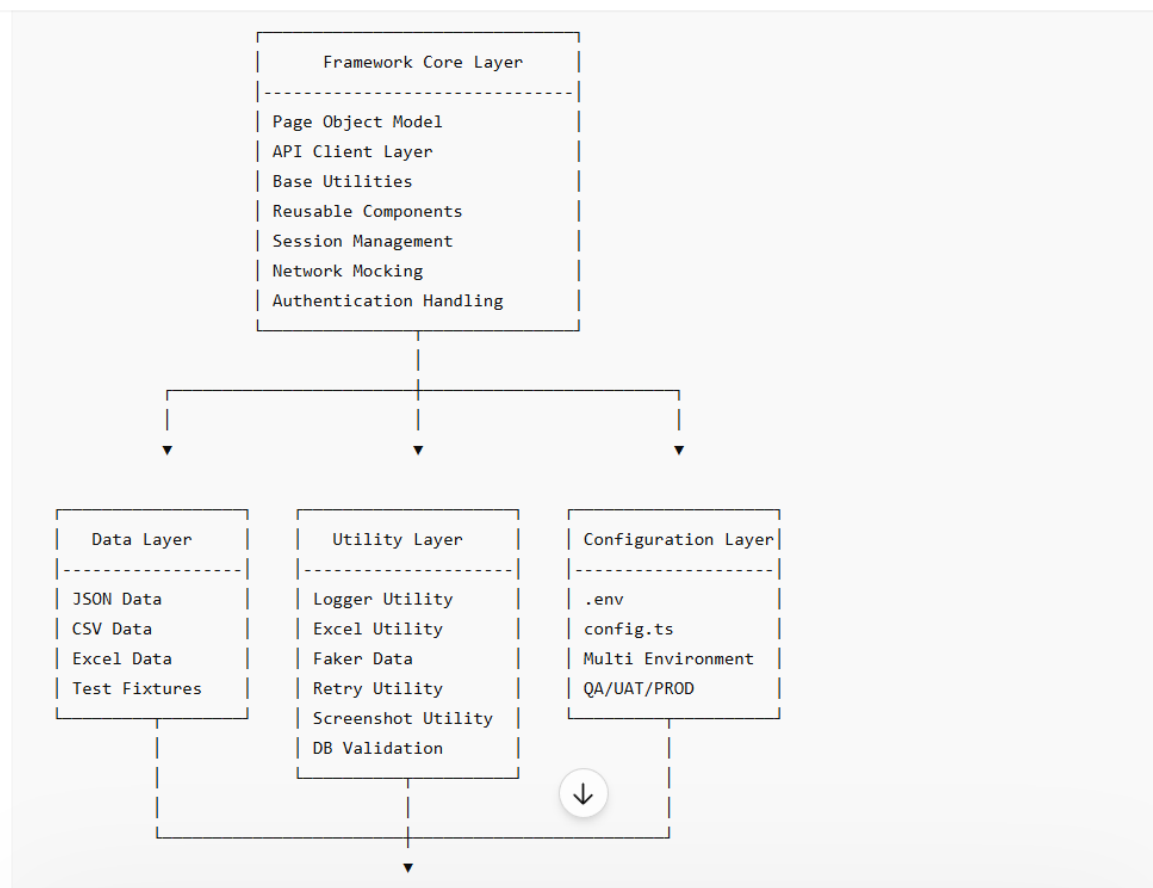
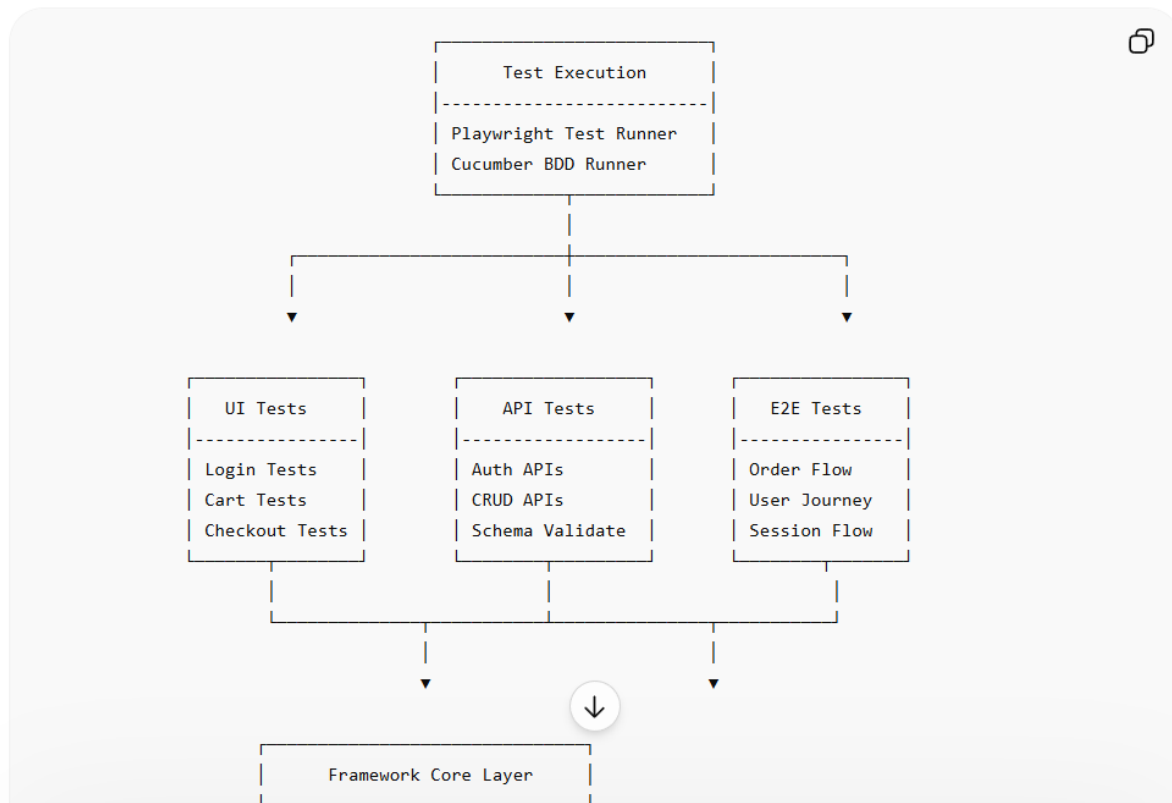
README.md

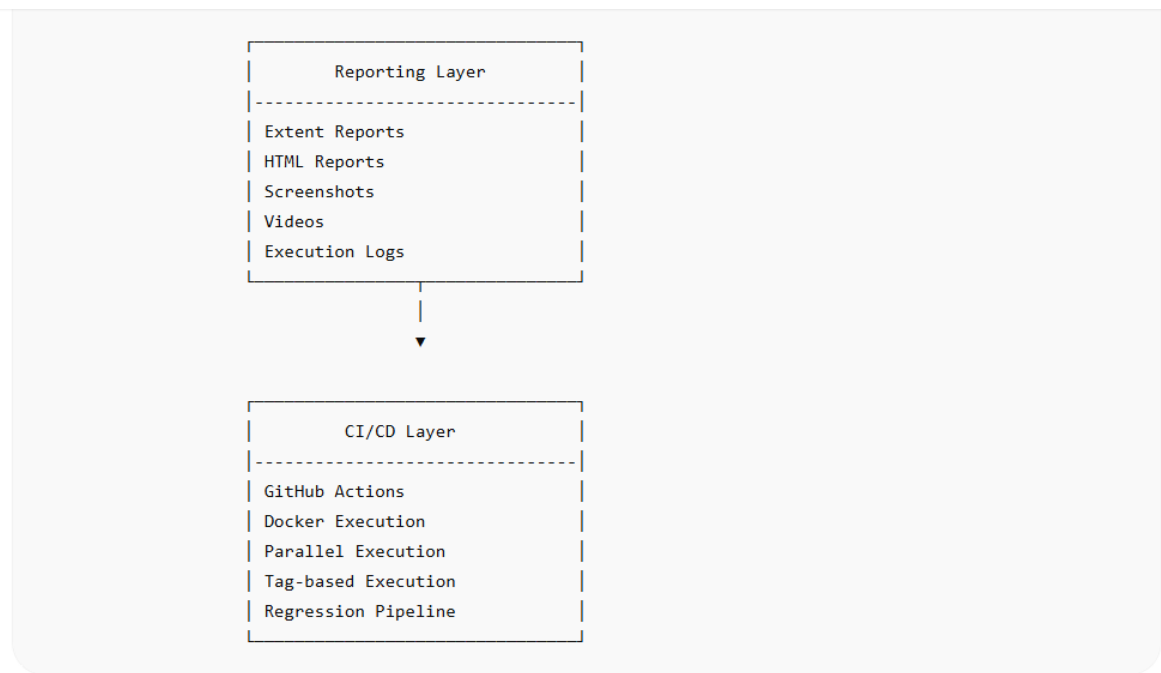
playwright-typescript-hybrid-framework.zip

Framework Highlights:

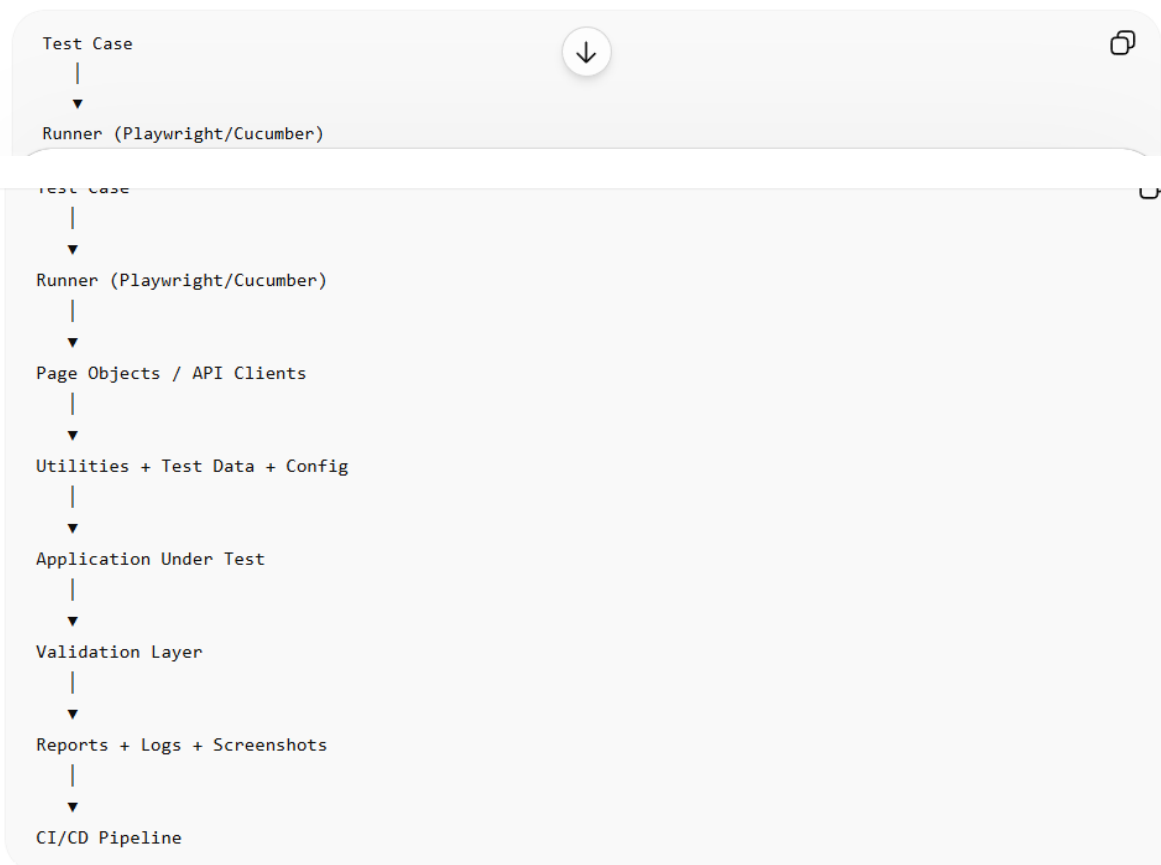
- Hybrid UI + API Automation
- Playwright + Cucumber support
- Authentication scenarios
- End-to-end test coverage
- Parallel & cross-browser execution
- Docker-ready setup
- GitHub Actions CI/CD
- Custom logging utility
- Data-driven framework support
- Extensible enterprise-level architecture

Playwright TypeScript Hybrid Automation Framework Architecture





High-Level Flow



Framework Design Patterns Used

- Hybrid Framework
- Page Object Model (POM)
- Data-Driven Framework
- BDD Framework

- Hybrid Framework
- Page Object Model (POM)
- Data-Driven Framework
- BDD Framework
- Utility-Based Reusable Architecture
- Environment-Based Configuration
- Layered Test Architecture

Key Enterprise Capabilities

- Cross-browser execution
- API + UI combined validation
- Authentication/session reuse
- Negative and edge case testing
- Dockerized execution
- Parallel execution
- Retry mechanisms
- Tag-based execution
- Extensible reporting
- CI/CD ready
- Database validation support
- Visual and accessibility testing ready

